# Ubuntu “Preparing Ubuntu” Error: Diagnosis and Solution

Diagnosis, Installation, Migration from Kubuntu to Ubuntu, and Final Cleanup

**Software and Hardware**

| | |
|---|---|
| **Type** | Technical Note |
| **Version** | V1 |
| **Status** | Final |
| **Last Updated** | 2026-03-20 |
| **Author** | Gabriel Gauna |
| **GitHub** | @gabygauna06 |
| **Instagram** | @gaby.gauna_ |

# Contents


**1 Introduction** **3**

**2 System and Initial Problem** **3**
2.1 Observed Behavior . . . . . 3

**3 Diagnosis of the Ubuntu Desktop Installer Failure** **4**
3.1 Inspection Commands . . . . . 4
3.2 Diagnostic Result . . . . . 4

**4 Change of Strategy** **5**

**5 Complete Removal of Windows and Disk Preparation** **5**
5.1 Path Used in HP BIOS . . . . . 5

**6 Kubuntu Installation** **5**
6.1 Partitioning Scheme . . . . . 6
6.2 Resulting State . . . . . 6

**7 Migration from Kubuntu to Ubuntu GNOME** **6**
7.1 Main Migration Commands . . . . . 6
7.2 Migration Verification . . . . . 7

**8 Initial Kubuntu Cleanup** **7**
8.1 Removal of Metapackages and the Former Display Manager . . . . . 7

**9 Graphical Environment Correction and GNOME Reinforcement** **8**
9.1 Commands Used . . . . . 8

**10 Ubuntu Terminal as Default Terminal Emulator** **8**

**11 Removal of KDE and Plasma Applications and Components** **9**
11.1 Removal of Visible KDE Applications . . . . . 9
11.2 Removal of Plasma Components . . . . . 9

**12 Final System Verification** **10**

**13 Command Summary** **10**
13.1 Installer Diagnosis . . . . . 11

13.2 Ubuntu GNOME Installation over Kubuntu . . . . . 11
13.3 Core Verification . . . . . 11
13.4 Initial Removal of Kubuntu Metapackages . . . . . 11
13.5 Ubuntu Theme and GNOME Reinforcement . . . . . 12
13.6 Default Terminal Configuration . . . . . 12
13.7 Removal of KDE Applications . . . . . 12
13.8 Removal of Plasma Components . . . . . 12
13.9 Final Cleanup . . . . . 13

**14 Additional Observations** **13**

**15 Conclusion** **13**

## Introduction

This document records the complete migration process of an HP OMEN 16 laptop from Windows 11 to a fully functional Linux environment ultimately consolidated as Ubuntu GNOME. The process was not direct or linear. Instead, it involved diagnosis of a failure in the official Ubuntu Desktop installer, adoption of an alternative installation path through Kubuntu, subsequent migration to the standard Ubuntu desktop environment, and gradual removal of residual KDE/Plasma components until the resulting system became practically equivalent to a clean Ubuntu installation.

The purpose of this note is to preserve the symptoms observed, the commands executed, the decisions made, and the final validation steps, so that the same procedure can be reused or adapted in the future if a similar situation occurs again.

**General Context**

The machine involved was an HP OMEN 16 with modern hardware, including a dedicated NVIDIA GPU. The original problem was not caused by the disk itself nor by Windows as such, but by the behavior of the Ubuntu Desktop graphical installer.

## System and Initial Problem

The machine used in this process was an HP OMEN 16 equipped with an AMD Ryzen 9 8940HX processor, 32 GB of DDR5 RAM, and an NVIDIA GeForce RTX 5060 Laptop GPU. The original goal was to install Ubuntu directly from a bootable USB drive.

The initial symptom was a freeze during the installation process at the screen labeled `Preparing Ubuntu`. At first sight, this could have been interpreted as a problem related to the live USB boot process, Windows 11, BitLocker, or storage detection. However, the observed behavior was more specific: the live environment itself did start correctly, but the graphical installer failed to proceed.

### Observed Behavior

The main behavior pattern was the following:

- the Ubuntu live USB booted successfully;
- the temporary live desktop loaded;
- when the installer was launched, the screen `Preparing Ubuntu` appeared;
- the process remained frozen indefinitely.

This made it necessary to distinguish between a general Linux boot failure and a failure specific to the installer.

## Diagnosis of the Ubuntu Desktop Installer Failure

In order to identify the actual cause of the problem, a terminal was opened inside the live session and the logs generated by the installer backend were examined. The Ubuntu Desktop installer uses `subiquity` as backend, so inspection of the files located in `/var/log/installer` made it possible to isolate the origin of the failure.

### Inspection Commands

```
sudo ls -lah /var/log/installer
sudo grep -RniE 'traceback|exception|fatal|error' /var/log/installer | tail -n 100
sudo tail -n 200 /var/log/installer/subiquity-server-debug.log
journalctl -b | grep -iE 'subiquity|curtin|traceback|error' | tail -n 120
```

### Diagnostic Result

The log analysis showed that the problem was not in disk detection or package installation. Instead, the backend crashed with a decoding error while attempting to process network-related information through `pyudev`. The main error reported was the following:

```
UnicodeDecodeError: 'utf-8' codec can't decode byte 0xec
in position 80: invalid continuation byte
```

The traceback also indicated a failure along the following chain:

```
subiquity -> probert -> pyudev -> ensure_unicode_string
```

In practical terms, this meant that the official Ubuntu Desktop installer crashed during the network discovery phase. As a consequence, the graphical frontend remained apparently stuck at `Preparing Ubuntu` even though the underlying issue was already a backend failure.

> **Diagnostic Conclusion**
> The problem was not Windows 11, not BitLocker by itself, not the NVMe drive, and not the live USB boot process. The failure was associated with the Ubuntu Desktop installer and its detection backend, specifically during processing of network-related information.

## Change of Strategy

Once the direct path through Ubuntu Desktop had been ruled out, the selected strategy was to use Kubuntu as an alternative flavor. The rationale was straightforward: Kubuntu shares the same Ubuntu base, but uses a different installer path, thereby avoiding the exact installer component that was failing in the Ubuntu Desktop variant.

This allowed the problem to be separated into two independent objectives:

- obtaining a functional Linux system on the machine;
- later converting that system into the preferred Ubuntu GNOME environment.

The priority therefore shifted from attempting to install Ubuntu directly to obtaining a stable Linux system first and transforming it into the desired desktop environment afterwards.

## Complete Removal of Windows and Disk Preparation

Since the decision was made to abandon Windows entirely, the dual-boot strategy was not pursued any further. Instead of shrinking partitions or disabling BitLocker in order to preserve Windows, a complete disk wipe was performed from the system firmware.

### Path Used in HP BIOS

The navigation sequence inside the firmware interface was:

```
Startup Menu -> BIOS Setup -> Security -> Secure Erase
```

Using `Secure Erase`, the contents of the 1 TB Samsung NVMe drive were completely removed. This left the storage in an unallocated state, ready for a new installation from scratch.

> **Important Warning**
> `Secure Erase` removes the previous operating system and all data stored on the drive. This procedure is only appropriate when Windows and all previous data no longer need to be preserved.

## Kubuntu Installation

With the disk already wiped, the installation process was restarted using a bootable Kubuntu USB drive. The Kubuntu installer loaded correctly and, unlike Ubuntu Desktop, successfully reached the language, installation mode, and partitioning screens.

### Partitioning Scheme

Because the disk had been completely erased, a manual EFI partition was created first, followed by a root partition for the operating system.

```
EFI System Partition:
- Size: 512 MB
- File system: FAT32
- Mount point: /boot/efi
- Flag: boot

Root partition:
- File system: ext4
- Mount point: /
- Size: remainder of the disk
```

This scheme was sufficient for a simple and clean installation, with no need to preserve any previous Windows partitions.

### Resulting State

The result of this stage was a functional Kubuntu system booting correctly on the laptop hardware. At this point, the primary problem had been solved: Linux was installed and usable.

## Migration from Kubuntu to Ubuntu GNOME

Once a stable base system had been achieved, the objective was no longer merely to have Linux running, but to transform the existing installation into an environment as close as possible to standard Ubuntu.

For that purpose, the main Ubuntu desktop packages and the corresponding display manager were installed.

### Main Migration Commands

```
sudo apt update
sudo apt upgrade -y
sudo apt install ubuntu-desktop gdm3
sudo ubuntu-drivers install
sudo reboot
```

During this process, `gdm3` was selected as the default display manager, replacing the typical Kubuntu login stack.

### Migration Verification

After rebooting, the state of the desktop, session manager, and NVIDIA driver was checked using:

```
nvidia-smi
echo $XDG_CURRENT_DESKTOP
systemctl status gdm --no-pager
```

The expected and effectively obtained result was:

- proper detection of the NVIDIA GPU and driver;
- desktop session reported as `ubuntu:GNOME`;
- `gdm` service active and running.

> **Successful Intermediate Result**
> At this stage the system was already fully usable as Ubuntu GNOME, although packages, themes, and applications from Kubuntu and KDE/Plasma were still present.

## Initial Kubuntu Cleanup

With Ubuntu GNOME already working, the first cleanup phase focused on removing global Kubuntu components. Before performing actual purges, simulations with `apt -s purge` were used to ensure that no critical GNOME packages would be removed.

### Removal of Metapackages and the Former Display Manager

```
sudo apt-mark manual ubuntu-desktop ubuntu-session gdm3
apt -s purge kubuntu-desktop kubuntu-settings-desktop sddm
```

Once the simulation confirmed that the removal was safe, the actual commands were executed:

```
sudo apt purge kubuntu-desktop kubuntu-settings-desktop sddm
sudo apt autoremove --purge
sudo reboot
```

This step significantly reduced the direct Kubuntu footprint, although visual traces and some Plasma components still remained.

## Graphical Environment Correction and GNOME Reinforcement

After the first cleanup stage, some hybrid behavior remained: Kubuntu branding during boot, coexistence of Ubuntu and KDE visual elements, and a mixture of KDE applications with the GNOME desktop.

To reinforce the Ubuntu environment, visual Ubuntu packages were reinstalled and the Yaru theme family was explicitly re-applied.

### Commands Used

```
sudo apt install --reinstall ubuntu-desktop ubuntu-session gdm3 gnome-shell \
gnome-control-center nautilus gnome-terminal yaru-theme-gtk yaru-theme-icon \
yaru-theme-sound ubuntu-wallpapers plymouth-theme-spinner
```

The theme state was then corrected through:

```
gsettings set org.gnome.desktop.interface gtk-theme 'Yaru'
gsettings set org.gnome.desktop.interface icon-theme 'Yaru'
gsettings set org.gnome.desktop.interface cursor-theme 'Yaru'
gsettings set org.gnome.desktop.wm.preferences theme 'Yaru'
gsettings reset org.gnome.shell favorite-apps
```

Boot branding and related components were also regenerated:

```
sudo update-grub
sudo update-initramfs -u
sudo reboot
```

## Ubuntu Terminal as Default Terminal Emulator

One of the specific goals of the migration was to replace Konsole with GNOME Terminal as the default terminal emulator. This was achieved by installing the appropriate package and updating the alternatives system:

```
sudo apt install gnome-terminal
sudo update-alternatives --config x-terminal-emulator
```

Subsequent verification confirmed that the system was using:

```
/usr/bin/gnome-terminal.wrapper
```

as the default terminal emulator, thereby aligning the interactive experience with standard Ubuntu behavior.

## Removal of KDE and Plasma Applications and Components

After the main desktop migration had succeeded, additional cleanup targeted visible KDE applications and later the deeper Plasma stack. The criterion applied throughout this phase was always the same: simulate first, execute only after confirmation.

### Removal of Visible KDE Applications

Among the packages removed in this stage were:

```
kdeconnect
keditbookmarks
ktexteditor-katepart
```

using:

```
sudo apt purge kdeconnect keditbookmarks ktexteditor-katepart
sudo apt autoremove --purge
```

This solved, among other issues, repeated crashes associated with `kdeconnectd`, which were still appearing in the GNOME-based environment.

### Removal of Plasma Components

Finally, major Plasma components that no longer made sense in a system oriented to Ubuntu GNOME were removed:

```
sudo apt purge plasma-desktop plasma-desktop-data plasma-workspace \
plasma-workspace-data plasma-workspace-wallpapers plasma-discover \
plasma-discover-common plasma-discover-backend-fwupd \
plasma-discover-backend-snap plasma-browser-integration plasma-disks \
plasma-firewall plasma-nm plasma-pa plasma-systemmonitor \
plasma-thunderbolt plasma-vault plasma-integration \
kdeplasma-addons-data

sudo apt autoremove --purge
sudo reboot
```

After this stage, Kubuntu and Plasma applications were no longer part of the visible daily workflow, although some residual libraries remained without practical impact.

## Final System Verification

Final verification confirmed that the system had been successfully consolidated as Ubuntu GNOME. The main checks used were:

```
echo $XDG_CURRENT_DESKTOP
readlink -f /usr/bin/x-terminal-emulator
dpkg -l | grep -E 'plasma|kdeconnect|konsole|dolphin|kate|gwenview|ark|elisa|haruna'
```

The final verifiable state was the following:

- active desktop session: `ubuntu:GNOME`;
- default terminal emulator: `gnome-terminal.wrapper`;
- absence of visible Kubuntu applications in normal use;
- properly working NVIDIA driver;
- functional `gdm` display manager;
- stable operation after multiple reboots and several days of use.

## Command Summary

For future reference, the most relevant command groups used during the process are summarized below.

## Installer Diagnosis

```
sudo ls -lah /var/log/installer
sudo grep -RniE 'traceback|exception|fatal|error' /var/log/installer | tail -n 100
sudo tail -n 200 /var/log/installer/subiquity-server-debug.log
journalctl -b | grep -iE 'subiquity|curtin|traceback|error' | tail -n 120
```

## Ubuntu GNOME Installation over Kubuntu

```
sudo apt update
sudo apt upgrade -y
sudo apt install ubuntu-desktop gdm3
sudo ubuntu-drivers install
sudo reboot
```

## Core Verification

```
nvidia-smi
echo $XDG_CURRENT_DESKTOP
systemctl status gdm --no-pager
```

## Initial Removal of Kubuntu Metapackages

```
sudo apt-mark manual ubuntu-desktop ubuntu-session gdm3
apt -s purge kubuntu-desktop kubuntu-settings-desktop sddm
sudo apt purge kubuntu-desktop kubuntu-settings-desktop sddm
sudo apt autoremove --purge
sudo reboot
```

## Ubuntu Theme and GNOME Reinforcement

```
sudo apt install --reinstall ubuntu-desktop ubuntu-session gdm3 gnome-shell \
gnome-control-center nautilus gnome-terminal yaru-theme-gtk yaru-theme-icon \
yaru-theme-sound ubuntu-wallpapers plymouth-theme-spinner

gsettings set org.gnome.desktop.interface gtk-theme 'Yaru'
gsettings set org.gnome.desktop.interface icon-theme 'Yaru'
gsettings set org.gnome.desktop.interface cursor-theme 'Yaru'
gsettings set org.gnome.desktop.wm.preferences theme 'Yaru'
gsettings reset org.gnome.shell favorite-apps

sudo update-grub
sudo update-initramfs -u
sudo reboot
```

## Default Terminal Configuration

```
sudo apt install gnome-terminal
sudo update-alternatives --config x-terminal-emulator
readlink -f /usr/bin/x-terminal-emulator
```

## Removal of KDE Applications

```
sudo apt purge kdeconnect keditbookmarks ktexteditor-katepart
sudo apt autoremove --purge
```

## Removal of Plasma Components

```
sudo apt purge plasma-desktop plasma-desktop-data plasma-workspace \
plasma-workspace-data plasma-workspace-wallpapers plasma-discover \
plasma-discover-common plasma-discover-backend-fwupd \
plasma-discover-backend-snap plasma-browser-integration plasma-disks \
plasma-firewall plasma-nm plasma-pa plasma-systemmonitor \
plasma-thunderbolt plasma-vault plasma-integration \
kdeplasma-addons-data

sudo apt autoremove --purge
sudo reboot
```

### Final Cleanup

```
sudo apt update
sudo apt full-upgrade -y
sudo apt autoremove --purge -y
sudo apt purge ?config-files -y
```

## Additional Observations

An important observation during log inspection was the possible exposure of sensitive network information. Installer logs may contain network names, identifiers, and, in some cases, credentials or configuration details that should not be shared publicly. Whenever such information is exposed during troubleshooting, credentials should be changed and the incident should be noted.

## Conclusion

The complete process demonstrated that a failure in the official Ubuntu Desktop installer does not necessarily imply a general incompatibility between the hardware and Linux. In this case, the problem was confined to the installer and its detection backend, whereas the hardware itself proved perfectly capable of running Linux once an alternative installation path had been selected.

The effective strategy consisted in first installing a functional flavor, then migrating the system to the desired desktop environment, and finally removing the remnants of the original desktop stack. This methodology made it possible to transform an initially Kubuntu-based installation into a practical end state equivalent to Ubuntu GNOME, fully operational and suitable for daily use.

> **Practical Lesson**
> When a direct installation path fails, insisting on the same installer is not always the best solution. In many cases it is more efficient to install a compatible base system through another flavor and then reorient the environment toward the desired desktop.

## References

[1] Canonical. *Ubuntu Desktop Guide*. Ubuntu Documentation. Available at: https://documentation.ubuntu.com/desktop/. Accessed: 2026-03-20.

[2] Kubuntu Team. *Kubuntu Documentation*. Kubuntu Documentation. Available at: https://kubuntu-docs.readthedocs.io/. Accessed: 2026-03-20.

[3] Ubuntu Community Help Wiki. *UEFI Booting*. Ubuntu Community Documentation. Available at: https://help.ubuntu.com/community/UEFIBooting. Accessed: 2026-03-20.

[4] HP Support. *HP Secure Erase*. HP Customer Support Documentation. Available at: https://support.hp.com/. Accessed: 2026-03-20.

[5] Debian and Ubuntu Developers. *APT Manual Pages*. Linux Manual Pages. Available at: https://manpages.ubuntu.com/. Accessed: 2026-03-20.